Q. 55—What should be done if side-rod or back-pin breaks?

A.—Take off all broken parts, also side rod on opposite side of engine.

Q. 56—Can all four-wheel switch-engines be run with their own steam with the side-rods down?

A.—No; on some builds of engines the forward crank-pin is liable to strike cross-head or the key through piston-rod, as when side-rods are down crank-pin does not always pass the cross-head at the exact place where it will clear, as it must do when side-rods are working. Cut off the end of this key so it will clear, if that is all that is in the way. On some engines the eccentrics are not on the same axle the main rods are coupled to; these engines must be towed in if all side-rods are off.

Q. 57—Why do you take side-rods down on the opposite side to the broken one?

A.—To avoid straining or bending the rods or pins. If forward wheel slips when rod was on center some damage would be done.

Q. 58—What is the effect of sanding the rail while the engine is slipping without first shutting off steam?

A.—If an engine catches on sand while slipping, it is liable to spring a side-rod, break a crank-pin or spring the axle. The size of drivers has something to do with this; it is worse with a large wheel than with a very small one, like a "Consolidation" has.

Q. 59—Is it good policy to allow sand to run from one pipe only?

A.—No; it brings most all the strain on one side, while the power is coming to both sides of engine, and is likely to spring the axle.

Q. 60—How do you block up an engine for a broken driving spring or hanger?

A.—If engine was raised with jacks, would block up the end of equalizer that had been connected to broken part, so it was a little higher than before, to allow for setting. It is customary to also block up between driving-box and frame at the box where spring is broken. If this is a forward box, it puts the load on that box, which may be too much; it is better to block up over back driving-box, whichever spring is broken; the weight is carried there best. (See question 72.) If engine was raised by running up on blocks or wedges, would put a block on top of box under broken spring first, if possible, run that wheel up on wedge till the engine was raised up so equalizer could be blocked up level again; then put block over back box also, to carry what weight of engine the spring still at work on that side would not hold up; take out the broken spring or hanger if necessary. If equalizer is under frame and boxes, block under the end that will hold it in proper place.

Q. 61—With a broken equalizer?

A.—If on a standard eight-wheel engine do the same work as for broken driving-spring on that side. Take out broken parts, if necessary. If an engine equalizer, block on top of truck oil-boxes and under top bar of engine-truck frame. If it is the cross-equalizer on a four-wheel switch-engine, block up between top of forward boxes and engine-frame; some of these equalizers are located under the bottom rail of frame, with the hangers going up outside of frame, in which case you can block between hanger and frame. For broken cross-equalizer between the forward drivers of a mogul, it will be necessary to block on top of forward driving-boxes; if equalizer going to center pin is broken or disabled, a block can be put over cross-equalizer and under boiler, and thus get under the used of forward driving-springs.

Q. 62—With broken engine-truck spring or hanger?

A.—If it is a four-wheel engine-truck, block over the equalizers and under top bar of engine-truck frame close to band of spring, high enough so engine will ride level with other side; with mogul, over the truck-box. If engine-truck center-casting breaks on a standard engine, block across under truck-frame and center-casting and over the equalizers, from one side to the other; a couple of pieces of rail 4½ to 5 feet long come handy for this. Or you can put a solid block under the engine-frame next to cylinder-saddle and on top of truck-frame on each side. This plan will give you the use of the engine-truck springs, although it does not always hold the center-casting up against male casting under smoke-arch, so engine will track straight.

Q. 63—With broken intermediate equalizer on mogul?

A.—Block over driving-boxes if necessary, as with the cross-equalizer broken; under the boiler and over cross-equalizer if engine-truck equalizer is disabled.

Q. 64—With broken engine-truck center pin on mogul what is to be done?

A.—Block up same as for broken equalizer, except that a block is needed over truck-axle and under front end of equalizer; a truck-brass comes handy for this purpose.

Q. 65—What should you do when a tire breaks and comes off the wheel on a standard engine?

A.—If it is a main tire, raise that wheel-center up off the rail a little higher than the thickness of the tire to allow for engine settling when blocked up; take out oil-cellar, so journal would not get cut on the edges of cellars; put a solid block of wood between pedestal-brace and journal to hold wheel-center up clear of rail; and block up over back driving-box, so engine could not settle or get down to allow cast-iron wheel-center to strike the rail. It will take considerable strain off the pedestal-brace to put a block under spring-saddle and on top of frame. Taking out this driving spring makes a sure job. Take off all other broken or disabled parts; if rods are still in good order, leave them up. If a back tire, block up in the same manner as for main tire, except that blocking comes next other journals and boxes. If engine is very heavy, it may be necessary to carry part of the weight of back end of the engine on tender. This can sometimes be done by wedging up under chafing-block on engine-deck and over coupling-bar; at other times it may be necessary to lay a solid tie or short rail on top of deck, the end against the fire-box, extending back into tender. Chain around this tie or rail and to the frame of back driving-box pedestal, and block up under end that is on tender, so weight of engine will be carried on rail or tie back of tender. (See questions 70 and 72.) This plan of blocking leaves three good tires on the rails, and the disabled wheel carried away from the rail. Run wheel on blocks to raise it clear of rail when possible.

Q. 66—With front tire on mogul or ten-wheel engine?

A.—Block up under journal of disabled wheel same as described in previous answer; in addition, it will be necessary to block up to put more weight on engine-trucks.

Q. 67—With main tire on mogul?

A.—Block up under main journal and over back driving-box. If with either tire broken on mogul or ten-wheel engine side-rods have to be taken off, it may be necessary to be towed in if crank-pin in forward wheel does not clear cross-head when side-rods are uncoupled. Some mogul and ten-wheel engines have the main tires without flanges, others have the forward pair "bald," which makes a little difference in keeping them on track when blocked up. (See question 56.)

Q. 68—With the back tire on mogul?

A.—Same as for back tire on any other engine, taking off all broken parts. To hold flanges of the good tire against the rail when running, chain from end of engine-frame and deck (the step-casting is handy for this) across to corner of tender behind the good tire; this will hold flange over and tender will be used to hold back end of engine on rail.

Q. 69—With both back tires on mogul?

A.—Raise both wheel-centers up to clear the rail and block under journals to hold them up. Arrange to carry part of the weight of back part of engine on tender, as per answer to question 65; chain back end of engine each way to tender-frame, so main wheels will have no chance to get off track. Or a shoe or "slipper" having a flange on one side can be fastened to wheel-center (a piece of old tire will make a good one), the wheel-center blocked so it will slide, and bring engine in that way. Another way is to take out the back wheels, as in case of a broken axle, and put in a car-truck, blocking up under engine-deck; this is a job for the wrecking-car. With a four-wheel switch-engine with front tire broken, if engine is still on track, front end of engine can be chained to a